capturing meaningful patterns that generalize to new data, resulting in overfitting. Label memorization is known to occur due to multiple factors such as complex, ambiguous features, and noisy labels [Baldock et al., 2021], which makes it difficult for the model to learn any meaningful relationship.

In this work, we specifically focus on introducing *noisy labels* as a way to study memorization, where we change the label of a particular class sample to a randomly chosen class label that is different from its original label. To ensure that the noisy label samples are memorized, we train the model until it achieves 100% training accuracy. Throughout our experiments, we introduce random label noise in all datasets by modifying 1% of the training set labels, maintaining consistency across evaluations.

3.2 Investigating LayerNorm (LN) Impact on Memorization and Learning

Removing LN parameters. To examine the role of LayerNorm (LN) in memorization and learning within transformers, we analyze the effects of omitting its learnable parameters, during training. This provides insights into how LN influences the balance between learning and memorization for both Pre- and Post-LN models. We precisely analyze the impact of LN on memorization and learning in Sec. 4. Please note that we use *LN removal* and *LN parameters removal* interchangeably in our paper. They both refer to removal of learnable parameters of the LN layer, while keeping the normalization operation, $N(x)$, intact.

Effect of LN Removal across Layers. To further understand the impact of LN at different stages of the model, we categorize the layers into - early, middle, and later layers (described in detail in Appendix F.4). We then selectively remove LN parameters from one set at a time to analyze how their absence affects learning and memorization. This analysis reveals which set of LNs is the most influential towards memorization and learning behaviors in Pre- and Post-LN transformers. The experiments and results are discussed in Sec. 5.

LN Gradients Analysis across Layers. To support our observations on the influence of LN, we compute the gradient of the loss function ($\mathcal{L}$) with respect to the input of LN, x, represented as $\nabla_x \mathcal{L}$ or $g_x = \frac{\partial \mathcal{L}}{\partial x}$. This measure quantifies how much the input to the LN layer affects the model's loss, thereby its learning and memorization ability.

To understand the sensitivity of each layer's LN towards memorization and learning, we compute the L2-norm of this gradient (i.e., $\|g_x\|_2$). Specifically, to quantify sensitivity towards *learning*, i.e., how the model generalizes the patterns to the test set, we compute $\|g_x\|_2$ for every test-set sample and average it across all of them, obtaining ***learning gradient norm***, denoted by $\|g_x^{\text{learn}}\|_2$. For *memorization*, we compute $||g_x||_2$ for each of the *noisy labels* samples that we injected into the train set and then averaged across all such noisy samples to obtain ***memorization gradient norm***, denoted by $\|g_x^{\text{mem}}\|_2$.

A higher gradient norm indicates that the layer's LN significantly influences the model's ability to memorize or learn, while a lower gradient suggests minimal impact. The discussion of memorization and learning gradients and their significance is shown in Sec. 6.

3.3 Key Metrics: Learning Accuracy, Memorization, Recovery & Random Predictions Score

To evaluate the impact of LN on the learning and memorization ability of the transformer models, we focus on several key metrics that provide insights into their behavior and effectiveness in the presence of noisy labels during training.

Learning (Test) Accuracy (%) refers to the model's performance on the test set, depicting how well it generalizes the learnt relationships to unseen data, marking it as a core indicator of the model's learning progress ($\frac{\text{\#Correct predictions on test set}}{\text{\#Total test set samples}} \times 100$). A high learning accuracy signifies that the model has learned meaningful patterns and is able to generalize well to unseen data. On the contrary, a low learning accuracy depicts poor generalizability.

Memorization Score (%) serves as an indicator of the model's tendency to memorize noisy labels that are irrelevant or erroneous, rather than genuinely learning the true underlying relationships ($\frac{\text{\#Noisy label samples memorized}}{\text{\#Total noisy label samples}} \times 100$). A high memorization score indicates that the model has overfit the noisy labels, effectively "memorizing" them.

Recovery Score (%) is a crucial metric that helps in understanding the impact of LayerNorm (LN) on memorization of noisy labels. It measures the model's ability to recover the genuine, true labels

Table 1: Summary of the impact of LN layer in Pre- and Post-LN Models.

If LN Removed	Learning Intact?	Memorization Mitigated?	Recovery Happens?
Pre-LN Model	✗ *Learning Disrupted*	✗ *Memorization Still Present*	✗ *Negligible Recovery*
Post-LN Model	✓ *Stable Learning*	✓ *Memorization Mitigated*	✓ *Genuine Labels Inferred*

after the removal of LN parameters ($\frac{\text{\#Recovered noisy label samples as true labels}}{\text{\#Total noisy label samples}} \times 100$). A high recovery score indicates that the model can successfully recover the original, correct labels by suppressing memorization.

Random Prediction Score (%) measures the percentage of noisy label samples whose predictions were changed to random labels after the removal of LN parameters. These predicted random labels are neither genuine nor the noisy label ($\frac{\text{\#Random predictions of noisy label samples}}{\text{\#Total noisy label samples}} \times 100$). Although this is not ideal, it provides a complete picture of the impact of LN parameters removal and indicates the extent to which the model can recover the true labels. A high percentage of random predictions suggests that the model struggles to recover the true labels effectively.

3.4 Datasets & Models Used

We empirically verify all claims and show extensive results against both language and vision modalities, including 3 language and 3 vision classification datasets, and 7 Pre-LN and 6 Post-LN transformers architectures, as follows:
Datasets: CIFAR10 [Krizhevsky et al., 2009], NICO++ [Zhang et al., 2023], UTK-Face [Zhang et al., 2017], Emotions [Saravia et al., 2018], News [Okite97, 2024], and TweetTopic [Antypas et al., 2022]
Post-LN Models: BERT [Devlin et al., 2019], RoBERTa [Yinhan et al., 2019], DistilBERT [Sanh et al., 2019], DeBERTa [He et al., 2020], ELECTRA [Clark, 2020], and Longformer [Beltagy et al., 2020]
Pre-LN Models: ViT-B [Alexey, 2020], ViT-S [Assran et al., 2022], DeiT [Touvron et al., 2021], GPT2 [Radford et al., 2019], GPT-Neo [Black et al., 2022], Qwen2 [Yang et al., 2024], and RoBERTa-PreLayerNorm [Ott et al., 2019].

It needs to be acknowledged that only language modality is available for the Post-LN architecture in practice/literature. We provide a thorough discussion of the datasets, models, and training configurations in Appendix F. All our experiments are run across 3 random seeds.

4 Impact of LN on Memorization and Learning

In this section, we examine the distinct impact of Layer Normalization (LN) on memorization and learning in Pre-LN and Post-LN transformers. To assess its influence, we train two versions of the model — one with LN parameters removed and one with them intact — and compare their performance using learning accuracy, memorization, recovery, and random prediction scores.

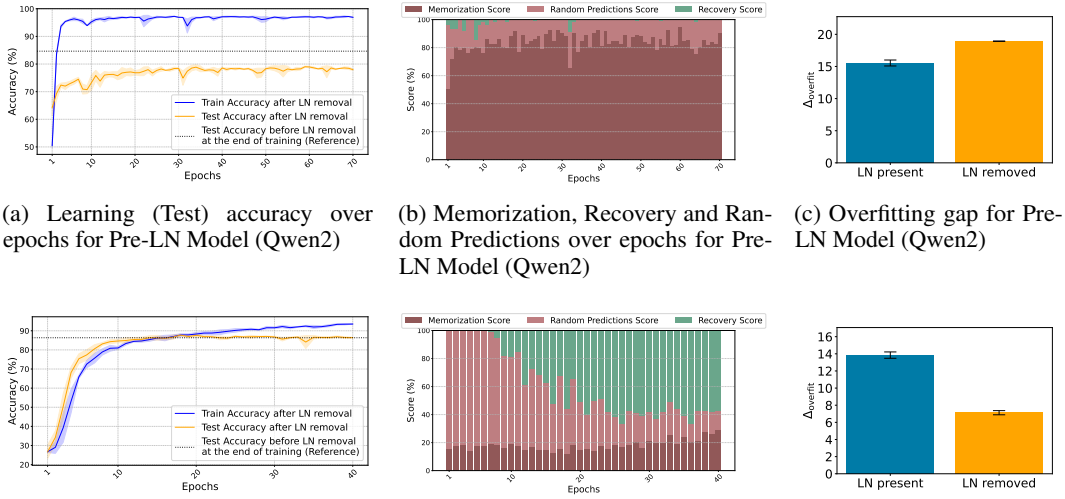
4.1 Learning Stability

From Figs. 1a & 1c, we observe that removing LN parameters in Pre-LN transformers significantly disrupts learning, whereas Post-LN transformers remain robust, maintaining their learning accuracy even after LN parameter removal.

This discrepancy becomes even more evident when analyzing the progression of learning in epochs, as depicted for Qwen2 (Pre-LN) in Fig. 2a & ELECTRA (Post-LN) in Fig. 2d. For Qwen2, once learning is disrupted by LN parameters removal, it does not recover till the end of training, indicating a fundamental instability. However, ELECTRA maintains stable learning throughout training, showing no signs of degradation, further highlighting its resilience to LN parameters removal. Similar results are observed for other Post-LN (BERT, DeBERTa, Longformer, RoBERT) and Pre-LN models (GPT2, GPTNeo, ViT-B, DeiT, ViT-S), as shown in Appendix G.1.

4.2 Memorization Suppression & Label Recovery

We now examine the role of LN in memorization and label recovery. From Figs. 1b & 1d, we observe that in Post-LN models, LN governs memorization, its parameter removal mitigates memorization



(a) Learning (Test) accuracy over epochs for Pre-LN Model (Qwen2)

(b) Memorization, Recovery and Random Predictions over epochs for Pre-LN Model (Qwen2)

(c) Overfitting gap for Pre-LN Model (Qwen2)

(d) Learning (Test) accuracy over epochs for Post-LN Model (ELECTRA)

(e) Memorization, Recovery and Random Predictions over epochs for Post-LN Model (ELECTRA)

(f) Overfitting gap for Post-LN Model (ELECTRA)

Figure 2: **LN removal destabilizes learning in Pre-LN models, while mitigates memorization in Post-LN models (News Dataset):** LN removal in Pre-LN models critically affects learning (accuracy gap in (a)) while Post-LN models remain robust (negligible gap in (d)); LN removal helps in effective mitigation of memorization and high recovery in Post-LN models (green bars in (e)), while memorization/random predictions still persist in Pre-LN models (red-color-family bars in (e)); LN removal in Pre-LN models exacerbates overfitting explained by increasing train-test accuracy gap in (c), and for Post-LN models it decreases due to memorization mitigation (see (f)).

and enhances true label recovery, reflected in lower memorization scores and higher recovery scores. In contrast, for Pre-LN models, LN parameters removal does not mitigate memorization, as indicated by persistently high memorization and random prediction scores.

This effect is even clearer when analyzing memorization over epochs, as shown for ELECTRA (Fig. 2e) and Qwen2 (Fig. 2b). In ELECTRA, memorization decreases over time, with label recovery improving as training progresses. Conversely, in Qwen2, memorization persists throughout training, and label recovery remains poor, indicating that LN parameter removal does not suppress memorization or aid label recovery in Pre-LN models. Similar patterns are observed in other Post-LN (BERT, DeBERTa, Longformer, RoBERT) and Pre-LN models (GPT2, GPTNeo, ViT-B, DeiT, ViT-S), as shown in Appendix G.1. These findings offer a nuanced perspective on LN's role: it is crucial for learning in Pre-LN models but does not influence memorization, contrary to prior work [Xu et al., 2019], which suggested that LN in Pre-LN models can contribute to overfitting.

In summary, **LN is essential for stable learning in Pre-LN models**, hence its parameters removal significantly destabilizes learning and widens the train-test accuracy gap ($\Delta^{\text{Pre}}_{\text{overfit}}$), i.e., exacerbating overfitting/memorization as illustrated in Fig. 2c. In contrast, in **Post-LN models, LN parameters removal suppresses memorization and enhances true label recovery**, thereby narrowing the train-test accuracy gap ($\Delta^{\text{Post}}_{\text{overfit}}$) as shown in Fig. 2f. This distinction is further illustrated in Table 1, which provides a comparative overview of LN's role in learning and memorization across Pre-LN and Post-LN model. Similar observations are observed in other Pre-LN (GPT2, GPTNeo, ViT-B, ViT-S, DeiT) and Post-LN (BERT, RoBERTa, DeBERTa, Longformer) models as reported in Appendix G.1.

5 The Pivotal Impact of LN in Early Layers

Building on the observation that LN has distinctive impacts on learning & memorization for Pre- and Post-LN models, respectively, we now precisely investigate the impact of the early, middle, and later LN layers on Pre- and Post-LN models. Fig. 3 depicts that early LNs are more significant than middle/later LNs in driving learning and memorization in both Pre- and Post-LN models.